Waterhose Teleop Scaling

Isaac-Waterhose-Robot-Demo-v0 | one-way split Newton runtime

Date: 2026-05-29   Branch: waterhose-demo   Commit: d54af20f1

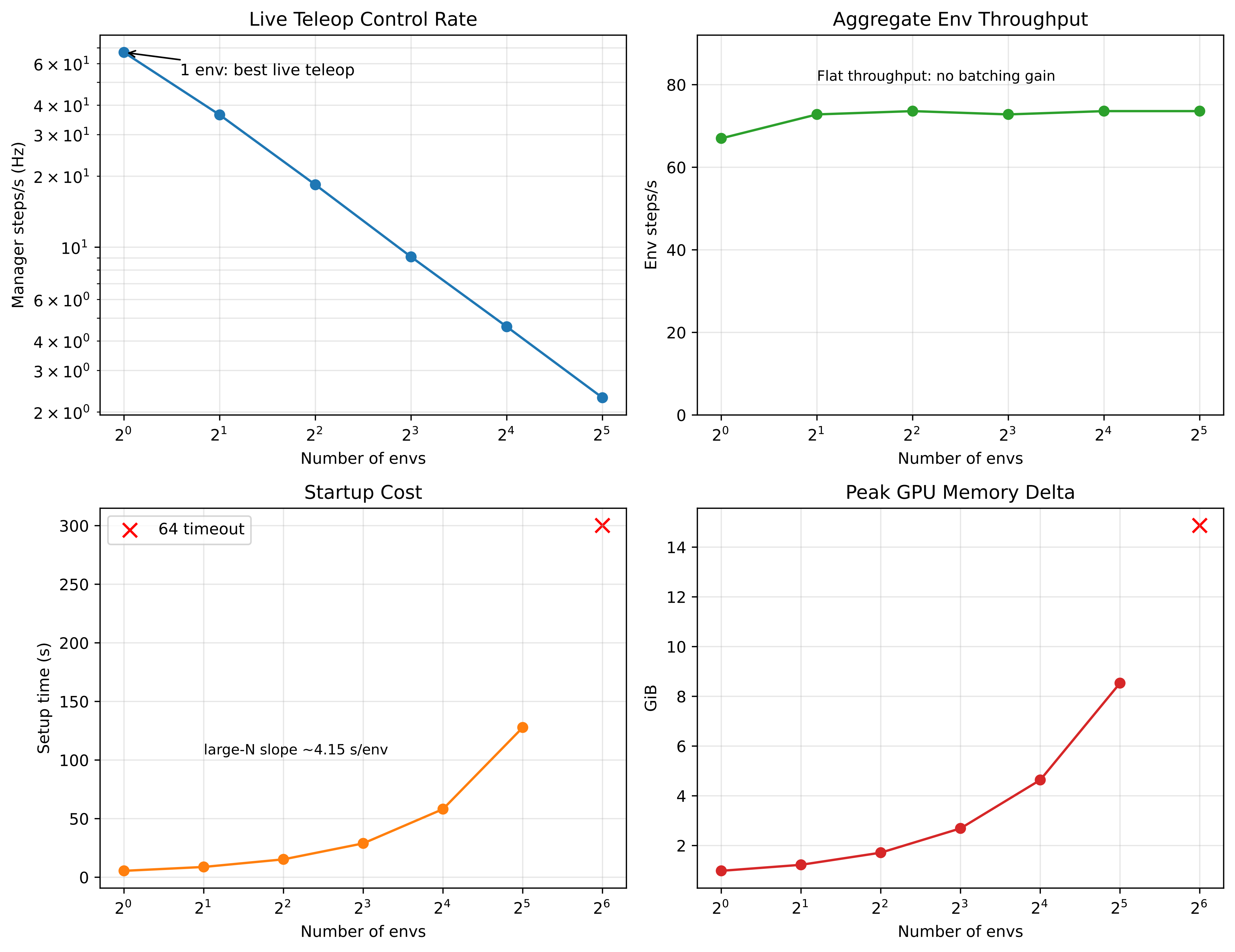
GPU: 0, NVIDIA GeForce RTX 5090, 32607 MiB

Executive summary: the current default waterhose task does not scale as a batched Isaac Lab environment. Aggregate env-step throughput remains nearly flat at about 73 env steps/s from 2 to 32 envs, while live teleop control rate falls inversely with num_envs. 64 envs did not complete within the 300 s timeout; higher requested points through 8192 were not run.

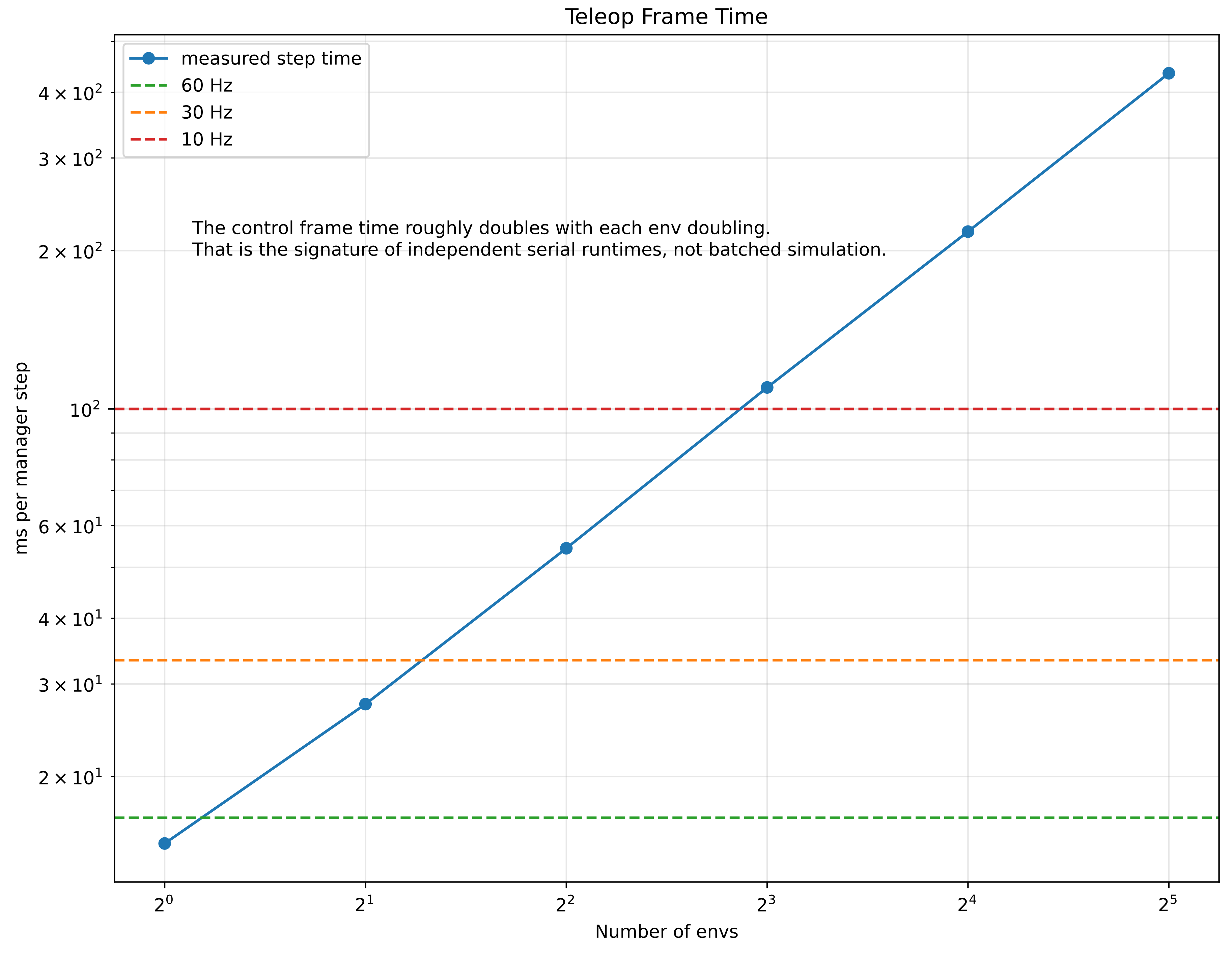
Methodology: each point launched run_robot_demo.py with --mode teleop, --teleop_device spacemouse, --vis none, --max_steps 60, and --profile. The SpaceMouse device was initialized normally; input was idle and broadcast to all environments. Live Hz is manager steps/s. Env steps/s is live Hz multiplied by num_envs.

Recommendation: use 1 env for live client teleop. 2 envs is borderline. 4-8 envs can be used for offline smoke checks, but they do not increase aggregate throughput. This architecture is not appropriate for high-throughput teleop data collection or training.

envs	status	setup s	live Hz	env steps/s	GPU GiB
1	completed	5.5	67.0	67.0	0.98
2	completed	8.8	36.4	72.8	1.23
4	completed	15.3	18.4	73.6	1.71
8	completed	28.9	9.1	72.8	2.69
16	completed	58.2	4.6	73.6	4.64
32	completed	127.8	2.3	73.6	8.53
64	timeout 300s	-	-	-	14.87



Live Teleop Control Rate
1 env: best live teleop
Manager steps/s (Hz)
Number of envs
Aggregate Env Throughput
Flat throughput: no batching gain
Env steps/s
Number of envs
Startup Cost
64 timeout
large-N slope ~4.15 s/env
Setup time (s)
Number of envs
Peak GPU Memory Delta
GiB
Number of envs


Teleop Frame Time
measured step time
60 Hz
30 Hz
10 Hz
The control frame time roughly doubles with each env doubling.
That is the signature of independent serial runtimes, not batched simulation.
ms per manager step
4×10^2
3×10^2
2×10^2
10^2
6×10^1
4×10^1
3×10^1
2×10^1
2^0
2^1
2^2
2^3
2^4
2^5
Number of envs